THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

December 18, 2015 9:30a.m. – 12:00noon

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL 6 questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (18 points)

In this question, you may use without proof the following if necessary:

- $\sqrt{2}$ is irrational.
- The product of any nonzero rational number and any irrational number is irrational.
- For any $n \in \mathbb{N}$, $\sqrt{n}$ is rational if and only if n is a perfect square.
- The sum/difference/product/quotient of two rational numbers is rational.

Prove or disprove each of the following statements:

(a) For any sets A and B, if $A \subseteq B$, then $A \times \mathcal{P}(A) \subseteq B \times \mathcal{P}(B)$.

(b) Let $\mathbb{I}$ be the set of irrational numbers. For any subset S of $\mathbb{I}$, if S is infinite, then S is uncountable.

(c) For any $m, n \in \mathbb{N}$ with $m > n$, if either $\sqrt{m}+\sqrt{n}$ or $\sqrt{m}-\sqrt{n}$ is rational, then both m and n are perfect squares.

2. (10 points)

A sequence $\{a_n\}$ is defined recursively by

$$a_1 = 6 \quad \text{and} \quad a_n = 5 - \frac{1}{4}a_{n-1} \quad \text{for any integer } n \geq 2.$$

(a) Prove by induction that

$$a_n = 2\left(\frac{-1}{4}\right)^{n-1} + 4 \quad \text{for any } n \in \mathbb{N}.$$

(b) Prove or disprove that $\{a_n\}$ is a convergent sequence.

3. (16 points)

Let R and S be relations on a nonempty set A. For each of the following statements, prove or disprove it when X is (i) $R \cup S$, (ii) $R \cap S$, (iii) $R \triangle S$.

(a) If R and S are reflexive, then so is X.

(b) If R and S are symmetric, then so is X.

(c) If R and S are transitive, then so is X.

4. (16 points)

Define $f : (0,1) \longrightarrow \mathbb{R}$ as follows:

$$f(x) = \begin{cases} \dfrac{1}{n-1} & \text{if } x = \dfrac{1}{n} \text{ for some integer } n \geq 2, \\ x & \text{otherwise.} \end{cases}$$

(a) Prove that f is injective.

(b) Find the range of f. Prove your answer.

(c) Construct a bijective function $g : [0,1) \longrightarrow [0,1]$. Prove your answer.

(d) Construct a bijective function $h : (0,1] \longrightarrow [0,1)$. Prove your answer.

(e) Use the results from (a) to (d) to prove that $(0,1)$ is numerically equivalent to $[0,1]$.

5. (24 points)

Let A and B be nonempty sets and let $f : A \longrightarrow B$ be a function. Consider the following statements:

(1) $B \setminus f(S) \subseteq f(A \setminus S)$ for any subset S of A.

(2) $f(A \setminus S) \subseteq B \setminus f(S)$ for any subset S of A.

(3) For any subsets S and T of A, if $f(S) = f(T)$, then $S = T$.

(4) For any subsets U and V of B, if $f^{-1}(U) = f^{-1}(V)$, then $U = V$.

Exactly two of the above statements are necessary and sufficient conditions for "f is injective" and the other two are necessary and sufficient conditions for "f is surjective". Make all the four assertions and prove each of them.

6. (16 points)

Let A and B be nonempty sets and let $f : A \longrightarrow B$ be a surjective function. For each $b \in B$, define

$$A_b = f^{-1}(\{b\}),$$

the preimage set of $\{b\}$ under f.

(a) Prove that A_b is nonempty for any $b \in B$.

(b) Let

$$P = \{A_b : b \in B\}.$$

Prove that P is a partition of A.

(c) Let $A = \mathbb{Z}$ and $B = \mathbb{Z}_2 \times \mathbb{Z}_3$, and let $f : \mathbb{Z} \longrightarrow \mathbb{Z}_2 \times \mathbb{Z}_3$ be defined by

$$f(a) = ([a]_2, [a]_3),$$

where $[a]_n$ denotes the equivalence class with respect to the congruence of integers modulo n.

(i) Prove that f is surjective.

(ii) Find the preimage set $f^{-1}(\{([b]_2, [c]_3)\})$ for each $([b]_2, [c]_3) \in \mathbb{Z}_2 \times \mathbb{Z}_3$. (*Remark*: There is no need to prove your answers.)

(iii) Use the above results to construct a partition of $\mathbb{Z}$.

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